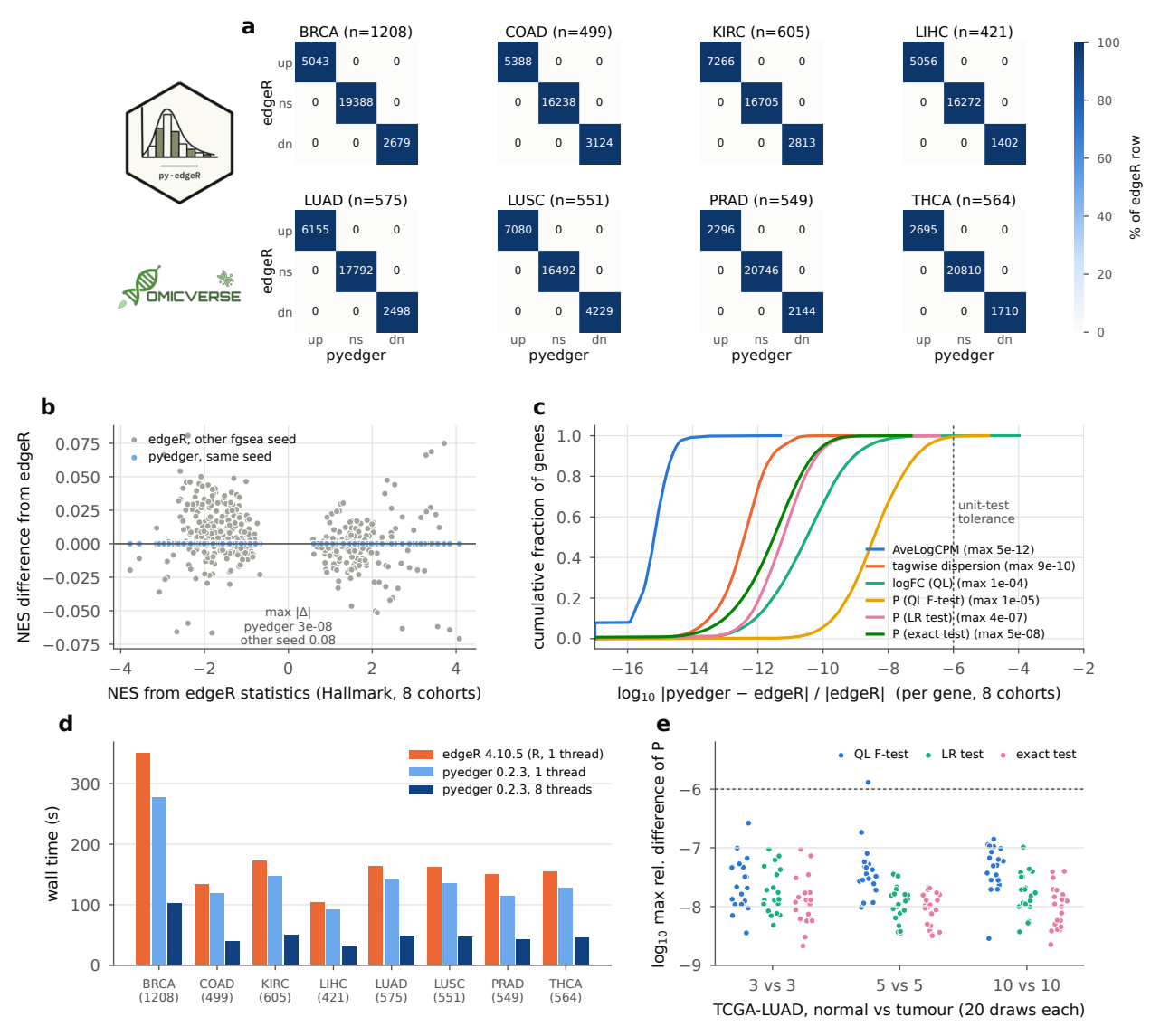

a
BRCA (n=1208)
COAD (n=499)
KIRC (n=605)
LIHC (n=421)
LUAD (n=575)
LUSC (n=551)
PRAD (n=549)
THCA (n=564)
edgeR
pyedger
up
ns
dn
% of edgeR row
b
edgeR, other fgsea seed
pyedger, same seed
NES difference from edgeR
NES from edgeR statistics (Hallmark, 8 cohorts)
max |Δ|
pyedger 3e-08
other seed 0.08
c
cumulative fraction of genes
unit-test tolerance
AveLogCPM (max 5e-12)
tagwise dispersion (max 9e-10)
logFC (QL) (max 1e-04)
P (QL F-test) (max 1e-05)
P (LR test) (max 4e-07)
P (exact test) (max 5e-08)
log10 |pyedger − edgeR| / |edgeR| (per gene, 8 cohorts)
d
wall time (s)
edgeR 4.10.5 (R, 1 thread)
pyedger 0.2.3, 1 thread
pyedger 0.2.3, 8 threads
BRCA (1208)
COAD (499)
KIRC (605)
LIHC (421)
LUAD (575)
LUSC (551)
PRAD (549)
THCA (564)
e
QL F-test
LR test
exact test
log10 max rel. difference of P
3 vs 3
5 vs 5
10 vs 10
TCGA-LUAD, normal vs tumour (20 draws each)